

WS4 — The formal hardness frontier

Where decision technology dies, how far each certificate class actually reaches, and what the reach is worth

1. What this work stream was for, and what it turned into

WS4 was planned as an assembly job: take known results about the undecidability of Collatz-like iteration, add our own theorems, and write down the boundary. The revision of July 27 changed the brief. By then the program had refuted three separate families of non-halting certificate, each measured in its own unit — DFA states, congruence modulus, backward depth — and the sharper deliverable was no longer the literature boundary but *our own*: three certificate families, each exact within its bound, each silent beyond it, and the growth constant of each bound.

Executing that produced four things the plan did not anticipate, and one of them is a correction to a claim this program has been making for two days.

(a) The units are not comparable, and when converted the headline is the weakest of the three. The LSB-first automatic-invariant bound — the result most of the WS1 effort bought — rules out congruence certificates at exactly 3 moduli. A direct sweep costing 30 seconds rules them out at 20,000. Section 5.

(b) The threshold argument I wrote first was unsound, and the fix made the theorem stronger rather than weaker. Section 5.2.

(c) A new proved lemma about machine 3 fell out of chasing the one modulus that survived: the parity of the next orbit value is the parity of the current 3-adic valuation. Section 5.3.

(d) The size frontier does not separate our cryptids from universal machines, and the plan's phrasing — "our cryptids sit strictly below" — presumed an answer nobody has. Section 3.4.

2. Measurement hygiene first

Everything in Section 6 is a ratio of solver times, so it is worth saying at the outset how those were obtained, because the earlier numbers in RESULTS.md were not obtained that way.

The searches that produced the impossibility bounds were run concurrently on one 10-core box at varying load, with a recorded uncertainty of roughly $\pm 30\%$. That is fine for the bounds themselves — a completed UNSAT is a completed UNSAT no matter how long it took — but it is not fine for a report whose content is the growth constants. The trigger was concrete: the MSB step from 12 to 13 states measured $10.18\times$ under heavy load and $5.47\times$ after two competing multi-day jobs were killed, and those two readings cannot be separated after the fact.

So the growth table here comes from `clean_growth.py`: one process, strictly sequential, nothing else of ours running, load average sampled at every instance and printed into the log. The effect is not subtle. MSB at $n = 10$ took 7.9 s here against 24.6 s in the loaded run. Every ratio below is between two timings taken under the same conditions.

3. WS4.1 — one syntax for cryptids and for universality

3.1 The definition

A **one-counter guarded affine machine** (GAM) has state an integer $x > 0$, a selector $\sigma(x)$ choosing among finitely many rule schemas, and rules $x \rightarrow (a_i x + b_i)/c_i$ guarded by exactness of the division. Two selector kinds occur in this project:

RES(d) — $\sigma(x)$ is the first rule whose guard divides x , so it is decided by $x \bmod d$ for d the lcm of the guards. Finitely many affine pieces.

VAL(q) — $\sigma(x) = v_q(x)$, the q -adic valuation. Infinitely many affine pieces, generated by one schema.

The point of writing it this way is that FRACTRAN is exactly RES with $b_i = 0$ and $a_i/c_i = p_i/q_i$. Conway's universality theorem is therefore a statement about this syntax and needs no translation layer — and Fenrir, our machine 7, is written in the very class Conway proved universal, which says nothing about Fenrir itself but means no translation is needed to compare them.

3.2 Register machines compile in, step for step

Construction (machine-verified). Encode a register machine state as $n = \prod p_j^{r_j} \cdot s_L$, with distinct primes for the registers and one prime per instruction label, the label prime appearing to the first power. Then

$$\begin{aligned} INC(j, k) \text{ at label } L &\rightarrow \text{the single fraction } p_j s_k / s_L \\ DEC(j, k, z) \text{ at label } L &\rightarrow s_k / (p_j s_L) \text{ then } s_z / s_L \end{aligned}$$

Exactly one label prime divides n at any time, so no two instructions can compete; within an instruction the decrement branch precedes the zero branch, which is what makes the guard order do the test. An I -instruction machine yields at most $2I$ rules, and the halt label has no rule, so the GAM halts exactly where the register machine does.

Verification. The compiled machine is run in lockstep with a direct register-machine interpreter. For $r0 += r1 \cdot r2$ the 6-instruction program compiles to 9 rules and agrees on all 1,072 steps across 64 starts, with every start also checked to produce the right answer — a program that silently loops cannot pass by vacuity. (gam.py.)

Consequence. With Minsky's theorem that register machines are Turing-complete, RES-GAMs are Turing-complete, and the smallest universal instruction count I gives an upper bound of $2I$ rules on where universality begins in our syntax. That is a real frontier statement, and it is an upper bound only.

3.3 An anchor that needs no citation

Conway's PRIMEGAME is a 14-rule RES-GAM with modulus 6,469,693,230. Run here from $x = 2$, the pure powers of two appearing in its orbit have exponents 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47 — the primes, consecutively, checked rather than quoted. So unbounded nontrivial computation is available in this syntax at 14 rules and one counter.

3.4 Where the cryptids sit — and what that does not prove

The Needle is one counter, VAL(2), and *one* rule schema: $F(x) = (2^{v+1} + 3)k + (2^v + v)$. Machine 3 is one counter, VAL(3), one schema. Written down, they are smaller than PRIMEGAME's 14 rules.

The plan said this meant they "sit strictly below" the universal machines. That does not follow, and I do not believe it. VAL(q) unfolds a single schema into infinitely many affine pieces — Section 4 shows the Needle really does have infinitely many distinct slopes — so smaller on the page is not smaller in power. Nothing in this section, and nothing we know of in the literature, lower-bounds the computational power of a one-schema VAL machine.

The open question this leaves, which is worth more than the section that produced it. Is the one-schema VAL(q) class Turing-complete? It is a genuine two-sided bet, the second the program has found and the first still open. Universality would show the size frontier is vacuous for our machines and would explain the resistance structurally. A decision procedure for the class would *decide our cryptids*. Both outcomes are consequential, which is exactly the property the program has learned to look for.

4. WS4.2 — the fault line, made checkable

The literature on one-dimensional piecewise-affine maps has a decidable island: reachability is decidable for injective maps with finitely many pieces cut out by intervals (LICS 2023). The plan asserted our machines sit outside it by branching type. Asserting is not enough for a hardness write-up, so both hypotheses are settled here by computation against the machines' own verified step functions.

4.1 The slope set is infinite

Proposition (proved; machine-verified). On the branch $v_2(x) = v$ the Needle map is affine with

$$F(x) = (1 + 3 \cdot 2^{-(v+1)})x + (v - 3/2)$$

and for machine 3, $G(a) = (1 + 3^{-(j+1)})a + (j + c_r - r/3)$ on $v_3(a) = j$. Checked against needle.step1 and machine 3's own accelerated step in exact rational arithmetic: 0 mismatches over $2 \leq x < 200,000$ (branches $v = 0..16$) and 0 over $2 \leq a < 200,000$ (branches $j = 0..10$).

The slopes are $5/2, 7/4, 11/8, 19/16, 35/32, 67/64, \dots \rightarrow 1$, pairwise distinct: all 400 of the first 400 are different, and the general form $(2^{v+1}+3)/2^{v+1}$ is visibly injective in v . A map with finitely many affine pieces has finitely many slopes. **So hypothesis (H1) fails, and no refinement of the partition can repair it** — the obstruction is in the slope set, not in how the pieces are described.

There is a second thing worth reading off this. The slopes tend to 1: the high-valuation branches are nearly the identity. The map is expanding on average while almost all of its individual pieces are almost neutral, which is why bounded-state reasoning about any fixed finite set of branches says nothing about the orbit.

4.2 The maps are not injective

$F(10) = F(12) = 17$: two different valuation branches land on one value. The witness is not exotic — 17 is the third element of the published orbit 6, 10, 17, 41, 101, ..., so the failure of injectivity is visible on the trajectory the whole problem is about. For machine 3, $G(16) = G(18) = 24$.

So (H2) fails as well, independently. The two hypotheses of the decidable island fail for unrelated reasons, which is a stronger statement than "our machines are not covered": there is no reformulation that sneaks in, because a reformulation would have to fix both.

5. WS4.3 — the certificate-impossibility account

5.1 In one variable, semilinear means congruence plus threshold

A subset of the naturals is Presburger-definable exactly when it is ultimately periodic: there are T and m with $x \in S$ iff $x + m \in S$ for all $x \geq T$. That is precisely the class $\{x < T\} \cup \{x \geq T : x \bmod m \in S\}$. Three literature strands collapse into this one class in our setting:

- **Linear-arithmetic non-termination certificates** — in one variable there is nothing else to write.
- **bbchallenge regular deciders** (FAR, WFAR, RepWL, CPS) — they operate on tapes encoding counters in unary or block form, and a regular language over a one-letter alphabet is ultimately periodic. What such a decider can say about a counter is exactly a congruence with a threshold.
- **The affine sieve** — whose only fuel is expansion in congruence quotients, provably absent here.

So refuting this one class refutes all three at once, and does so for a reason the bbchallenge wiki states only empirically.

5.2 The refutation, and the bug in my first version of it

The argument. Suppose $I = \{x < T\} \cup \{x \geq T : x \bmod m \in S\}$ contains the orbit and avoids the halting set H . Take an orbit element $x_i \geq T$ and an $h \in H$ with $h \geq T$ and $h \equiv x_i \pmod{m}$. Then $x_i \in I$ forces its class into S , so $h \in I$ — and $h \in H$. Contradiction. A certificate therefore exists only if no orbit element above T is congruent to any element of H above T .

The bug. My first implementation collected every residue that 2^e takes mod m and looked for a collision with those. That is unsound for the threshold claim. The sequence $2^e \bmod m$ is *eventually* periodic, and a residue in the pre-period is taken by finitely many powers of two — so a collision there is defeated by a large enough T . Mod 12, for instance, the powers of two run 1, 2, 4, 8, 4, 8, ...: a collision at residue 1 or 2 proves nothing about thresholds above 2.

The fix, which strengthened the theorem. Restricting to the eventual cycle makes every collision witness an *infinite* family of halting values, hence arbitrarily large ones, so the refutation holds for every threshold rather than for thresholds below some cutoff. The search got harder and the conclusion got better.

Result (machine-verified over the stated domain). For both machines, every modulus $2 \leq m \leq 20,000$ admits a collision at orbit index ≥ 500 , using the complete cycle residues of H . Hence **no congruence certificate with any modulus in that range and any threshold whatsoever**. The Needle's hardest modulus was 14,336, needing orbit index 11265; machine 3's was 13,122, needing index 105033.

The threshold-free half of this needs no computation at all and is worth recording separately. The Needle's b is strictly increasing (its T_2), so the orbit is unbounded; a certificate that is a finite union of intervals and contains an unbounded set must contain a ray $[T, \infty)$; every ray contains a power of two. **So no finite-union-of-intervals certificate exists, unconditionally** — the $m = 1$ case, proved

rather than swept.

5.3 A lemma found by chasing the last survivor

One modulus held out: $13,122 = 2 \cdot 3^8$ for machine 3, whose cycle residue is the single value 6561. A separating congruence would *prove* machine 3 never halts from its start, so it was worth diagnosing rather than dismissing. The required residue means $v_3(a) \geq 8$ together with an odd 3-free part, and among the orbit's first 60,500 values all eight with $v_3 \geq 8$ had an even 3-free part. Eight for eight looks structural at $p = 1/256$ — if the parity were balanced.

Lemma (proved; machine-verified). It is not balanced:

$$G(a) \equiv v_3(a) \pmod{2}$$

Proof. $G(a) = (3^{j+1}+1)m + (r \cdot 3^j + j + c_r)$ with $j = v_3(a)$, $r \in \{1,2\}$, $c_1 = 3$, $c_2 = 4$. The first coefficient is even and 3^j is odd, so $G(a) \equiv r + j + c_r \pmod{2}$; $r + c_r$ is odd for both values of r , giving $G(a) \equiv j$. **QED**
Verified: 0 mismatches over 399,987 values $a < 400,000$.

So the 3-free part of a machine-3 orbit value is odd only about a quarter of the time, not half, and eight even ones in a row has probability 0.10 — unremarkable. The survivor was luck, not structure, and extending the orbit killed it at index 105,033. **The lemma is the durable output: a new proved fact about machine 3, found only because a null result was checked instead of accepted.**

5.4 Three families in one currency

The three refuted families are measured in units that cannot be compared as written, which makes it easy to mistake a large number in one unit for a strong result. Two of them can be converted into each other exactly. A congruence certificate is a union of residue classes mod m , and the canonical residue trackers are

$$\text{MSB-first: state} = x \bmod m, \delta(c, d) = (2c + d) \bmod m \rightarrow m \text{ states}$$

$$\text{LSB-first: state} = (x \text{ so far } \bmod m, 2^i \bmod m) \rightarrow m \cdot \text{ord}_m(2) \text{ states}$$

so "no automatic certificate at $\leq n$ states" rules out every modulus whose tracker fits in n states. The asymmetry is severe, and it is the same asymmetry the solver timings measured — here showing up as a statement about content rather than about seconds.

certificate family	unit of the bound	reach, in moduli	what it covers
congruence + threshold	modulus m	every $m \leq 20,000$, any threshold	unions of residue classes only
MSB automatic (≤ 13 states)	DFA states	moduli 2..13 (12 of them)	any 2-automatic set
LSB automatic (≤ 11 states)	DFA states	moduli 2, 3, 4 (3 of them)	any 2-automatic set
backward-depth counting (WS2)	depth L	exact counts at every fixed L	density, not separation

Read the right way round, this reorders the program. On congruences the 30-second sweep beats the SAT bounds by 1,538×. On everything else the SAT bounds are the only statement there

is, because a union of residue classes is a vanishing fraction of the automatic sets of any given size. Neither family contains the other, and the LSB bound — the headline of this program for two days, and the thing three multi-day jobs were spent on — is the weakest of the three on this axis, at 3 moduli.

That is not an argument that WS1 was wasted; it is an argument that its value was never in the state count. What WS1 actually established is the *shape* of the obstruction — that a certificate must avoid the whole halting basin, not just H — and that finding survives the conversion intact.

6. The growth constants, re-measured

Each family's bound is bought with compute, and the price per unit is the part of the account that tells you what not to buy next. Under the single-load protocol of Section 2:

n	MSB sec	MSB step	LSB sec	LSB step	LSB/MSB
6	—	—	0.7	—	—
7	—	—	3.1	4.30×	—
8	0.7	—	13.4	4.35×	18.98
9	1.9	2.73×	74.5	5.55×	38.62
10	7.9	4.10×	705.0	9.46×	89.23
11	54.6	6.91×	—	—	—
12	613.3	11.23×	—	—	—
13	8,046.7	13.12×	—	—	—

The rise is real, and it is far cleaner than the contaminated series suggested. Both encodings now come out *strictly monotone*: MSB at 2.73, 4.10, 6.91, 11.23, 13.12 and LSB at 4.30, 4.35, 5.55, 9.46. The loaded MSB measurement had wandered — 4.33 down to 3.46, up to 5.47, 6.70, 10.18, then back down to 5.47 — and every one of those reversals was load rather than mathematics.

Two claims settle here. **The earlier reading that the per-state factor had fallen back to 5.47 at $n = 13$ is withdrawn**; it arrived precisely as two competing multi-day jobs were killed, and under clean measurement that step is the largest in the series at 13.12×. And the claim that the last step is the largest in every series, which had rested on four noisy series, now rests on two clean ones and holds in both.

The cross-encoding ratio behaves the same way: 18.98, 38.62, 89.23 at $n = 8..10$. It compounds, so MSB carries a genuinely smaller per-state factor rather than a constant discount. Both arguments of the old exchange-rate formula $\log_g(C)$ drift, which is why no particular number from it survived; what survives is the order of the answer, one or two states.

The operational content is unchanged and now rests on clean numbers: buying more states is the worst-value move available. The next MSB state costs about 13× the last one, and the last one bought 12 moduli of congruence coverage that a 30-second script covers 1,538× better.

7. What WS4 changes about the program's own claims

claim	status after WS4
"Our cryptids sit strictly below the universal machines"	Withdrawn. The frontier is an upper bound on where universality starts. One-schema VAL(q) is not known to be weaker than a 14-rule RES machine, and Section 3.4 makes that a stated open question.
"No automatic certificate at ≤ 12 MSB states"	Now ≤ 13. The $n = 13$ run completed (UNSAT). This is the last state either encoding will buy at a sane price.
"The per-state factor fell back to 5.47 at $n = 13$ "	Withdrawn. That reading came from a window in which two competing jobs were killed. Under single-load measurement the MSB series is monotone rising.
"Our no-congruence theorems explain regular-decider resistance"	Upheld and quantified. One-variable semilinear = congruence + threshold, refuted to modulus 20,000 for any threshold.
"The LSB bound is the program's headline impossibility result"	Demoted. Converted to moduli it covers 3. Its durable content is the halting-basin shape, not the state count.

8. Traps this stream added

Convert bounds into a common currency before ranking them. Three numbers in three units invite the reader — and the author — to rank them by size. Two of ours differed by three orders of magnitude in the direction opposite to how they had been presented. The conversion took twenty lines.

When a sweep leaves survivors, diagnose them; do not widen the window until they vanish. The one modulus that held out was worth an hour because a genuine survivor would have been a proof. Diagnosing it produced a lemma; widening the window alone would have produced nothing but a bigger number.

An argument about "arbitrarily large" needs the residues taken infinitely often, not the residues taken. The pre-period is where this kind of threshold argument leaks, and the leak is invisible in the output: the sweep reported success either way.

Do not measure growth constants under variable load. Bounds survive contaminated timing; the constants that price them do not. Fixing the protocol changed a headline reading from non-monotone to monotone.

9. Status of every claim in this report

claim	epistemic status
Register machine $\rightarrow$ RES-GAM is step-exact, $\leq 2I$ rules	proved + machine-verified (lockstep, 2 programs, 100 starts)
RES-GAMs are Turing-complete	proved, <i>citing</i> Minsky; the compiler is ours and verified

claim	epistemic status
PRIMEGAME is a 14-rule GAM emitting the primes	machine-verified here (first 15 primes), not cited
One-schema VAL(q) is/is not universal	open — stated, not answered
Needle and machine 3 have infinite slope sets	proved (injective in the branch index) + machine-verified
Neither map is injective	proved by explicit witness
No finite-union-of-intervals certificate	proved (monotone orbit + every ray meets H)
No congruence certificate, $m \leq 20,000$, any threshold	machine-verified over the stated modulus range; the orbit side is a finite prefix, which only strengthens each refutation
$G(a) \equiv v_3(a) \pmod{2}$	proved + machine-verified (400,000 values)
No MSB automatic certificate at ≤ 13 states	machine-verified (completed UNSAT, cross-validated four ways)
Growth constants in Section 6	measured under one stated load condition; ratios within the run are comparable, cross-run comparisons to older logs are not

Code: *gam.py*, *branch_type.py*, *congruence.py*, *certificate_classes.py*, *clean_growth.py*, all with logs of record in the same directory. Report regenerated by *make_ws4_report.py*, which reads every number from those logs.